Powering SIESTA's Electronic Structure Calculations with the SIESTA ToolBox

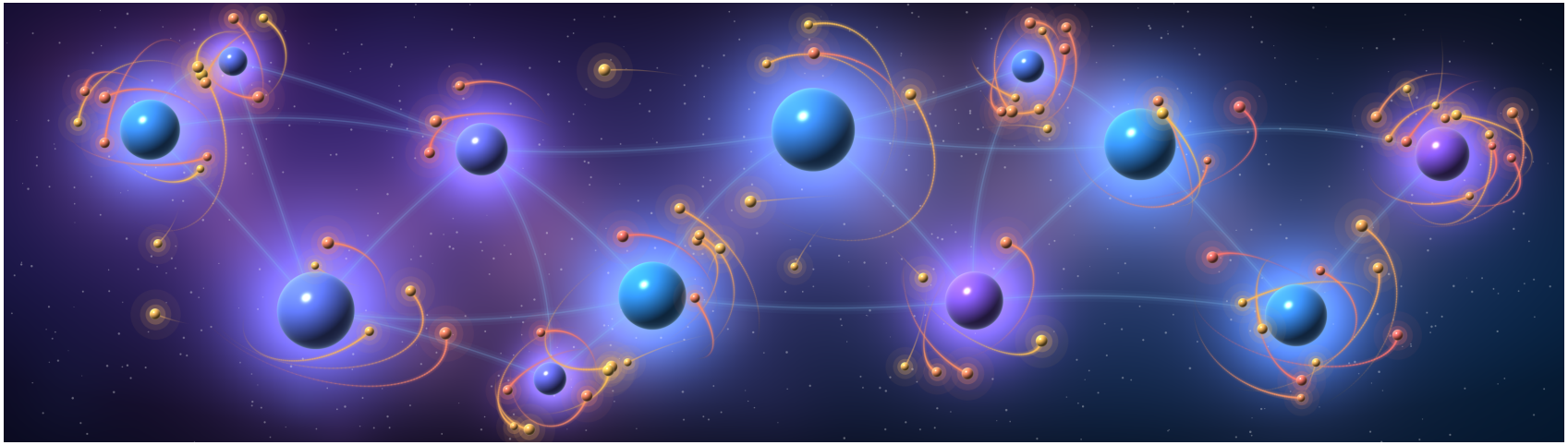
Dr. Carlos Maciel de Oliveira Bastos

International Center of Physics
University of Brasilia

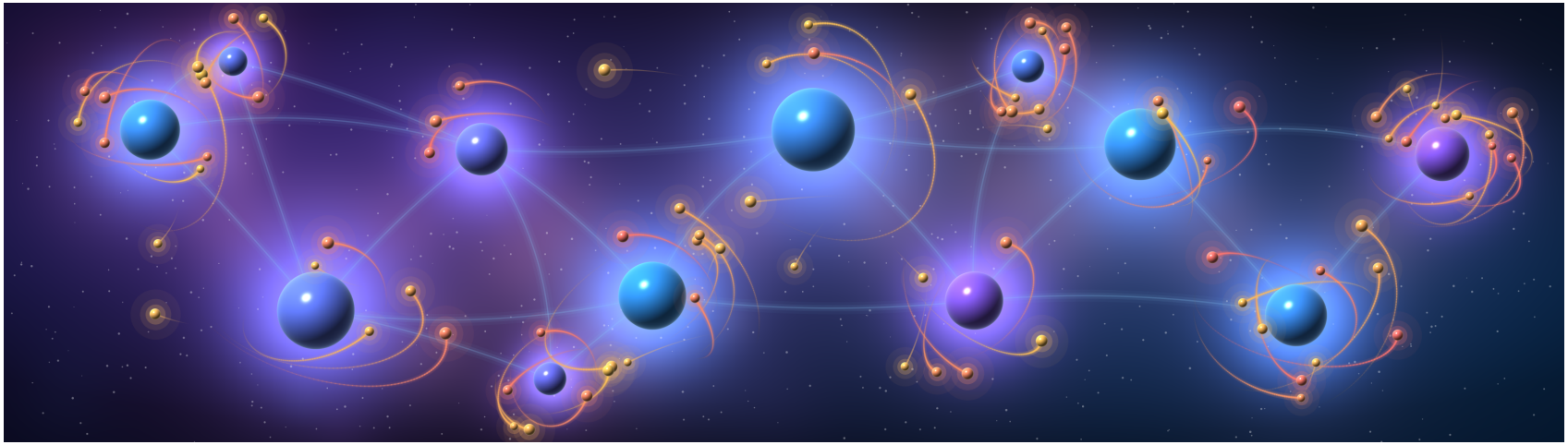


A Brief History of Density Functional Theory

The Many-Body Problem and the Hohenberg-Kohn Theorems



The Many-Body Problem and the Hohenberg-Kohn Theorems



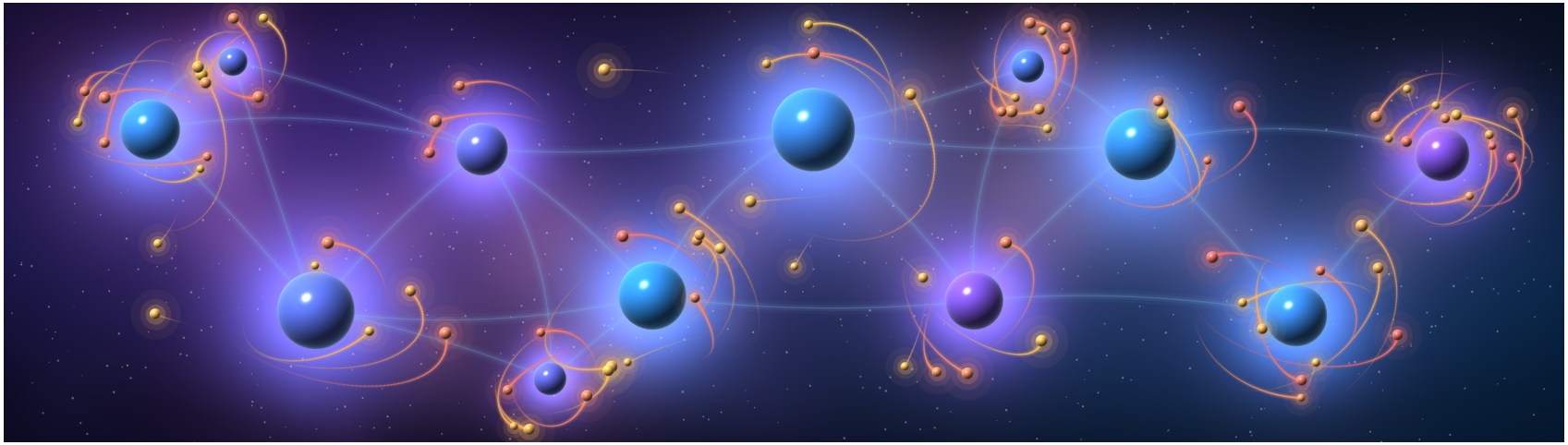
First Hohenberg-Kohn Theorem

The ground-state density $\rho_0(\mathbf{r})$ uniquely determines the external potential, and therefore the full many-body wave function:

$$|\psi_0\rangle = |\psi[\rho_0]\rangle$$

$3N$ variables are replaced by a function of 3 variables.

The Many-Body Problem and the Hohenberg-Kohn Theorems



First Hohenberg-Kohn Theorem

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$3N$ variables are replaced by a function of 3 variables.

Second Hohenberg-Kohn Theorem

The energy functional is variational: for any trial density $\rho' \neq \rho_0$,

$$E[\rho_0] < E[\rho'] \quad \Longleftrightarrow \quad E_0 = \min E[\rho]$$

This gives us a practical route to the ground state.

The Kohn-Sham Scheme and the Exchange-Correlation Problem



Electronic Density

- Interacting System
- Real System



Electronic Density

- Non-Interacting System
- Ficticius system



Electronic Density

- Non-Interacting System
- Ficticius system with the electronic density is the same of Real system

The Kohn-Sham Scheme and the Exchange-Correlation Problem



Electronic Density

- Interacting System
- Real System



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Electronic Density

- Non-Interacting System
- Ficticius system with the electronic density is the same of Real system

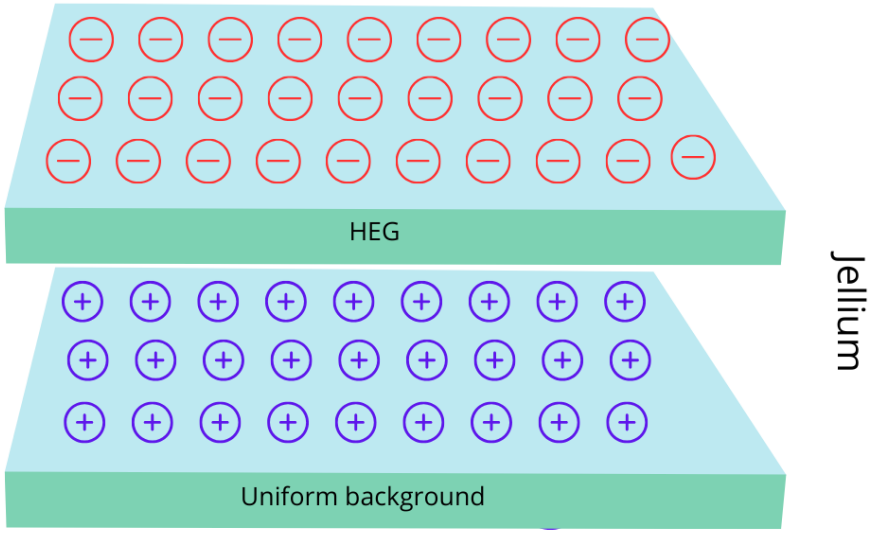
Kohn-Sham mapping

Replace the interacting system by an **auxiliary non-interacting system** with the *same* ground-state density:

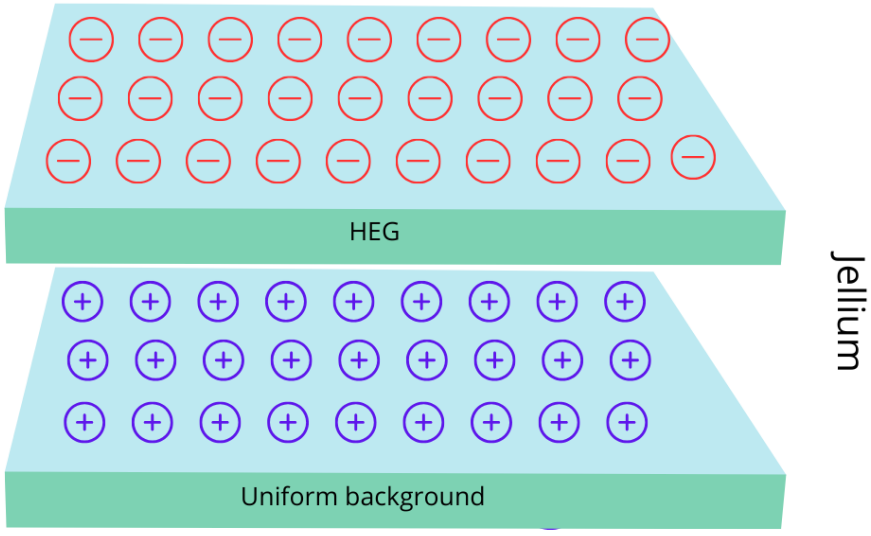
$$E[\rho] = T_s[\rho] + E_H[\rho] + E_{\text{xc}}[\rho] \qquad \{T_s + v_H[\rho] + v_{\text{xc}}[\rho]\}\,\phi_i^{\text{KS}} = \epsilon_i \phi_i^{\text{KS}}$$

All the difficulty is now contained in a single unknown term, $E_{\text{xc}}[\rho]$.

The Kohn-Sham Scheme and the Exchange-Correlation Problem



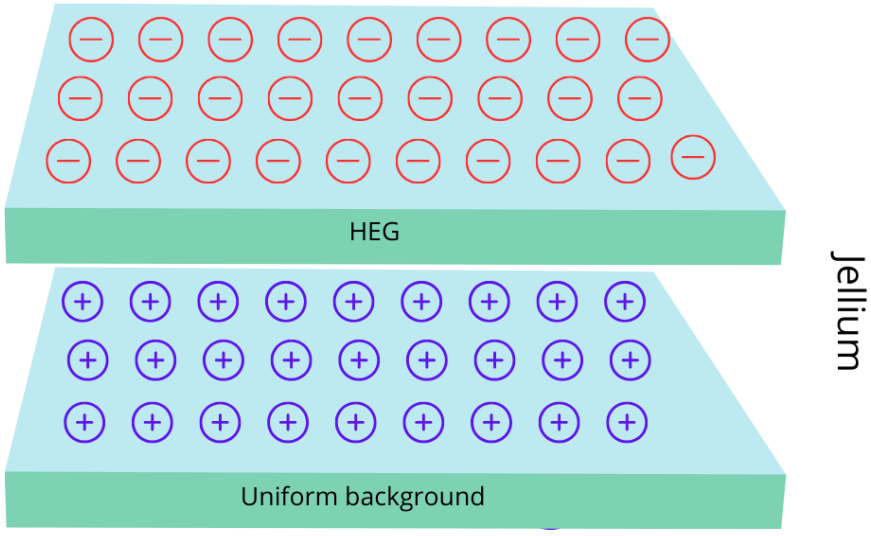
The Kohn-Sham Scheme and the Exchange-Correlation Problem



From jellium to real materials

- **LDA:** built from the homogeneous electron gas (jellium).
- **GGA (PBE):** adds the density gradient.
- **Hybrids (HSE06):** mix exact exchange, correcting the self-interaction error and improving band gaps.

The Kohn-Sham Scheme and the Exchange-Correlation Problem



From jellium to real materials

- **LDA:** built from the homogeneous electron gas (jellium).
- **GGA (PBE):** adds the density gradient.
- **Hybrids (HSE06):** mix exact exchange, correcting the self-interaction error and improving band gaps.

Long-range dispersion

- Semilocal functionals do not describe van der Waals interactions.
- **Grimme-type corrections (D3)** and non-local vdW functionals are essential for layered and porous systems.

A Brief History of Atomic Basis

Basis Sets and Numerical Atomic Orbitals

Choosing a basis

Basis functions represent electronic states, and their size and flexibility set the balance between accuracy and computational cost.

Plane waves

A widely used basis for periodic systems. Increasing the energy cutoff enlarges the basis and improves convergence.

Localized orbitals

Gaussian orbitals, numerical atomic orbitals, and wavelets can reduce cost and support linear-scaling methods.

SIESTA's numerical atomic orbitals (PAOs)

Generated from Kohn–Sham solutions for isolated pseudoatoms with the same pseudopotential and exchange-correlation functional. A confining potential makes them vanish beyond a cutoff radius r_c.

Single-zeta (SZ): the minimal basis, with one radial function per occupied atomic channel.

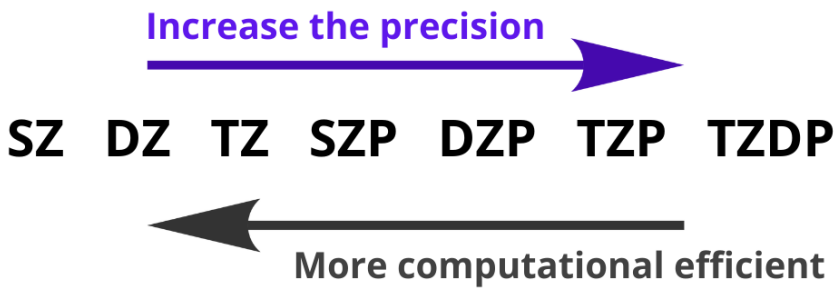
Building Flexibility: SZ, DZ, TZ, and Polarization

- **SZ:** one radial function per channel.
- **DZ/TZ:** add radial functions to improve flexibility.
- **Polarization (P):** add higher-angular-momentum orbitals.

Example: silicon

The illustrated basis grows from 4 orbitals (SZ) to 8 (DZ) and 13 (DZ+P).

Atom	Valence configuration	SZ		DZ		P	
		# orbitals	symmetry	# orbitals	symmetry	# orbitals	symmetry
Si	$3s^2\ 3p^2$	1	s	2	s	1	d_{xy}
		1	p_x	2	p_x	1	d_{yz}
		1	p_y	2	p_y	1	d_{zx}
		1	p_z	2	p_z	1	$d_{x^2-y^2}$
						1	$d_{3z^2-r^2}$
	Total	4		8		(DZ+P) 13	



Basis-Set Convergence and Cost

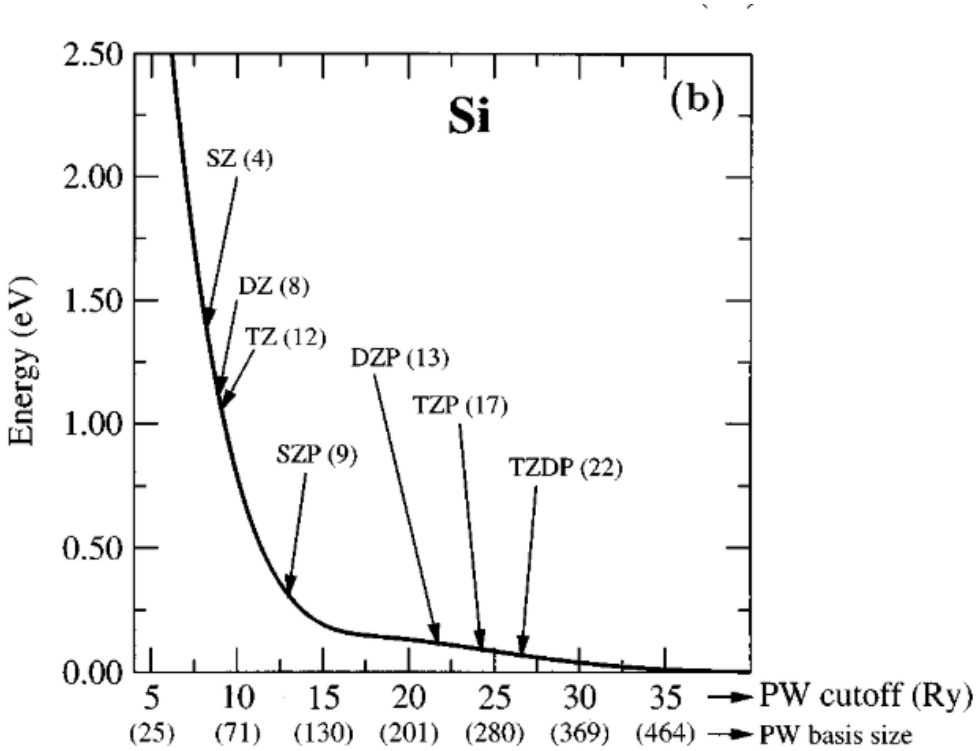
A silicon example

Increasing the plane-wave cutoff lowers the energy toward convergence, but also increases the number of basis functions.

Localized alternatives

SZ, DZ, TZ, and polarized NAOs offer a ladder of accuracy and computational cost.

Illustrative comparison; convergence must be checked for each system.



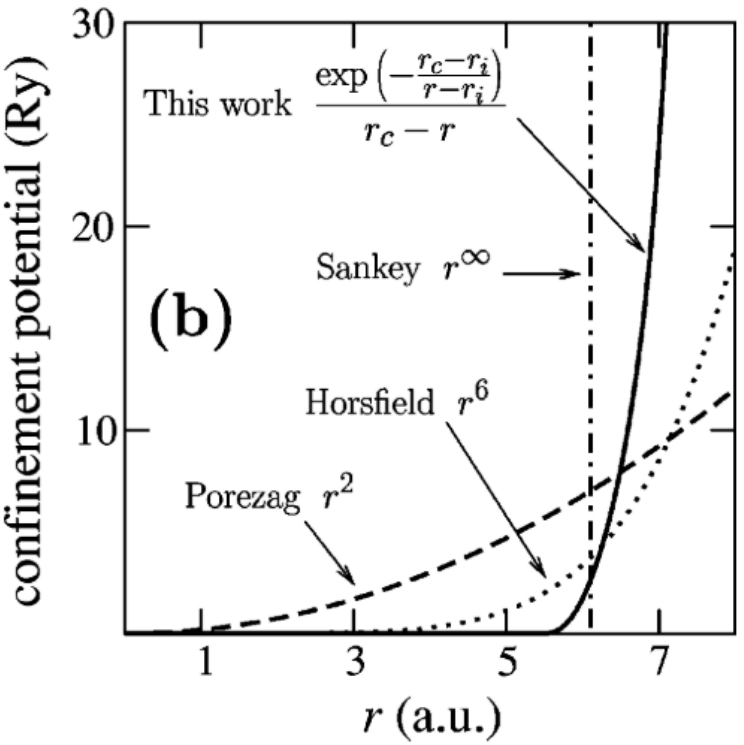
Cutoff Radius and `PAO.EnergyShift`

Confining the orbitals

A confining potential truncates each PAO at a finite radius r_c. In SIESTA, `PAO.EnergyShift` sets this range through the orbital's energy increase under confinement.

Practical trade-off

A larger energy shift generally produces more compact orbitals. Test convergence to balance locality, accuracy, and cost.



Examples of confining potentials for atomic orbitals.

A Brief History of Pseudopotentials

Why Pseudopotentials?

The core-electron problem

Core states are tightly bound and often do not change appreciably in chemical processes. Treating them explicitly adds computational cost.

Plane-wave calculations

Rapid oscillations near nuclei require many high-frequency plane waves, slowing convergence.

The idea

Replace the core region by an effective potential acting on valence electrons.

Smoother valence states need fewer plane waves while preserving the relevant physics outside the core.

From Orthogonal Plane Waves to Pseudopotentials

A brief history

Herring introduced orthogonal plane waves (OPWs) in 1940; in 1959, Phillips–Kleinman and others developed the pseudopotential approach.

Enforcing orthogonality to the core

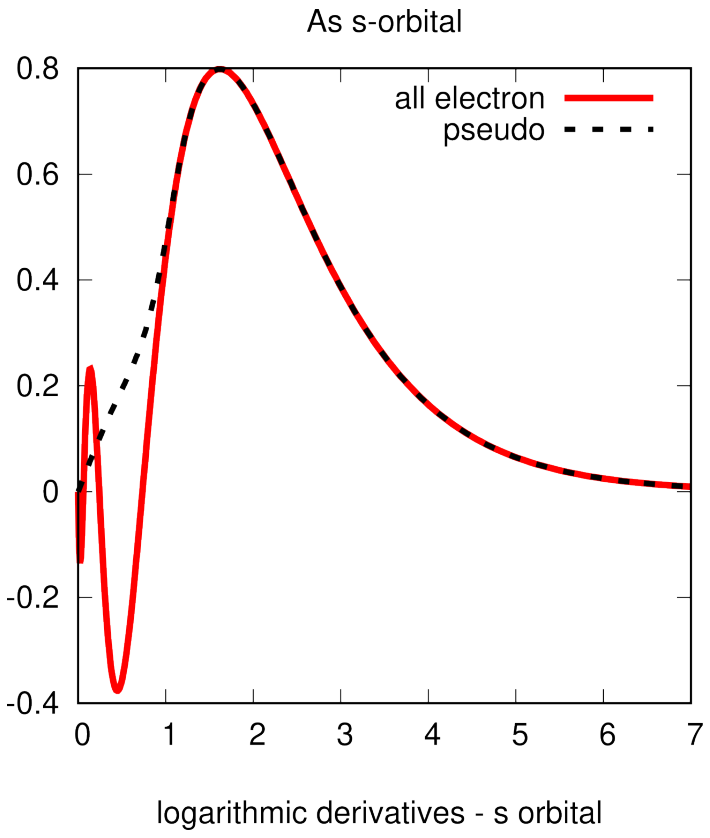
An OPW removes components along the core states using the projector $\hat{P}_c$:

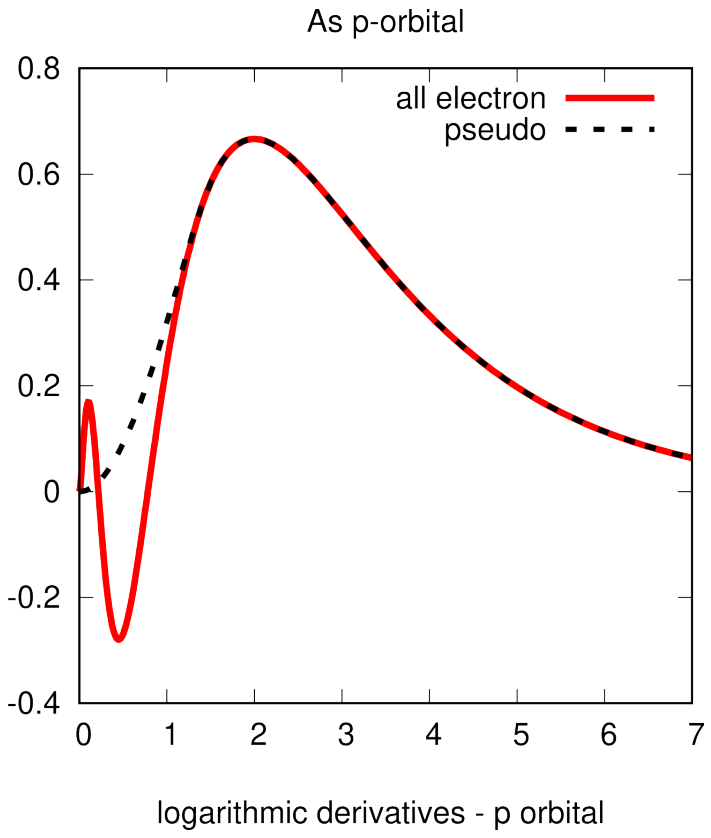
$$|\mathbf{k}+\mathbf{G}\rangle_{\mathrm{OPW}} = (1-\hat{P}_c)|\mathbf{k}+\mathbf{G}\rangle,$$
$$\hat{P}_c = \sum_c |\phi_c\rangle\langle\phi_c|.$$

This constraint can be recast as an effective repulsive contribution.

All-Electron vs Pseudopotential

Orthogonality to core states yields a repulsive correction, $\hat{V}_{\mathrm{PS}} = \hat{V} + \hat{V}_R$; smooth pseudo-states reproduce valence behavior outside the core radius.





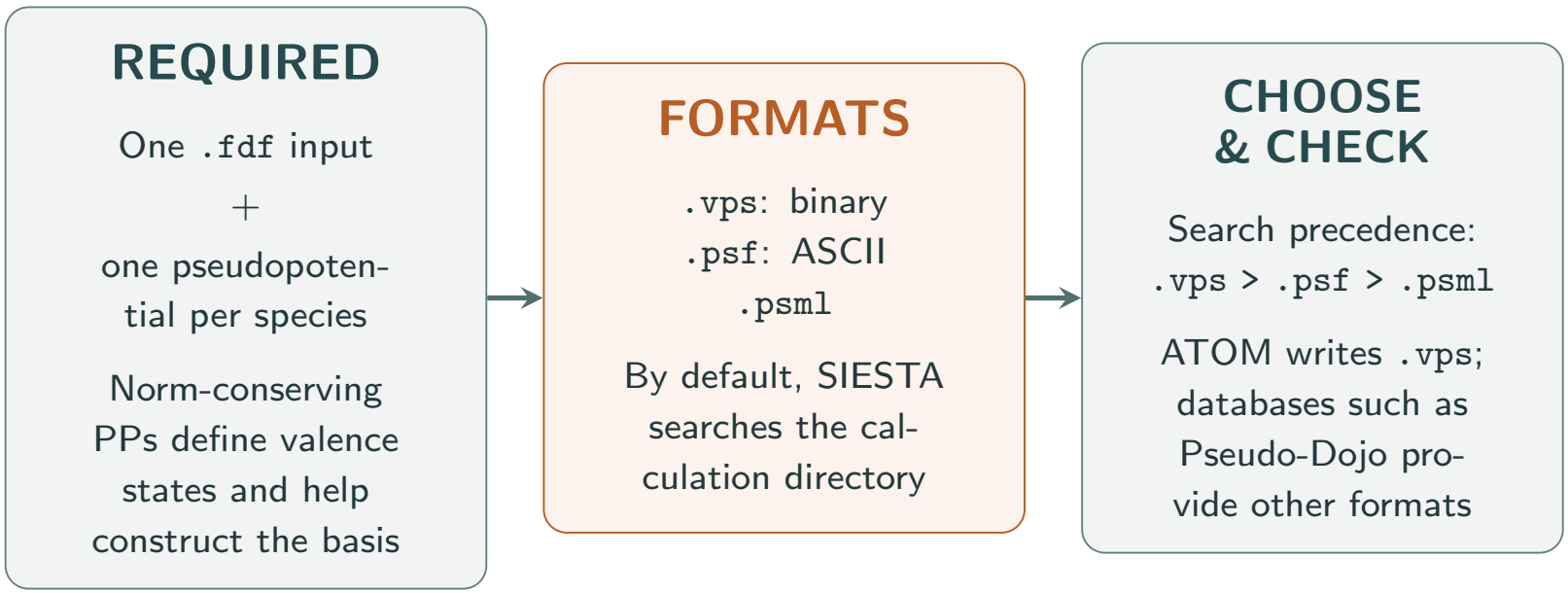
Siesta Implementation

What SIESTA Does

SIESTA simulates materials from first principles.

It uses density-functional theory (DFT) to compute electronic structure, total energies, and atomic forces. It also performs structural relaxation and molecular dynamics. Localized numerical atomic orbitals and norm-conserving pseudopotentials enable efficient simulations of large systems.

Input Files and Pseudopotentials



Before production
Validate pseudopotential quality for the system and property of interest.

The FDF Input: Rules That Prevent Errors

01 Syntax

`#` starts comments. Labels ignore case; punctuation and spaces are generally ignored.

02 Values

Logicals: `T`, `true`, `.true.`, or a bare flag. Write real values with a decimal point.

03 Units & structure

Include units (e.g. `LatticeConstant 10 Ang`); use `%block`/`%endblock` and `%include`.

04 Debug

Check `fdf.log` for misspelled or unrecognized labels, and verify values filled from defaults.

Define the System and Its Species

SPECIES

`ChemicalSpecies Label`

Index → atomic number → label → pseudopotential

```
1   6   C   C.psf
```

ATOMS

`AtomicCoordinatesAnd AtomicSpecies`

Each row gives x, y, z and the species index

CONSISTENCY

`NumberOfSpecies` = species rows

`NumberOfAtoms` = coordinate rows

`SystemLabel` names outputs; `SystemName` describes the calculation

Choose a Basis for Accuracy and Cost

Localized numerical atomic orbitals — `PAO.BasisSize`

SZ

Single-zeta minimal basis

DZ

Double-zeta

SZP / DZP

Add polarization orbitals

`PAO.EnergyShift` controls confinement and cutoff radii
Default `0.01 Ry`; smaller shifts generally extend orbitals

Converge the Electronic-Structure Calculation

01 K-points

`Kgrid.MonkhorstPack`: e.g. 3×4 grid + offset; simplified form uses $M \times N \times P$. Respect cell symmetry.

02 XC & spin

Pair `XC.Functional` (LDA/GGA/VDW) with `XC.Authors` (e.g. PBE). Choose spin treatment; spin-orbit needs a relativistic PP.

03 Numerical controls

Converge `Mesh.Cutoff` for reliable energies and forces. `MaxSCFIterations` caps SCF cycles per MD step.

04 Validate

Test k-point sampling, mesh cutoff, and basis size against the target observable; do not trust defaults blindly.

SIESTA Output Files

ELECTRONIC STATES

`.EIG`: eigenvalues
`.bands`: band structure
`.DOS`: total DOS
`.PDOS`: projected DOS

STRUCTURE & FORCES

`.XV`: final coordinates/cell
`.FA`: final-step forces
`.ion.xml`: ion/restart data

MOLECULAR DYNAMICS

`.MD`: trajectory
Atomic positions and velocities

REAL-SPACE FIELDS

`.RHO`: charge density
`.VT`: total potential
`.VH`: Hartree potential
`.LDOS`: local DOS

ORBITALS & MATRIX

`.WFSX`: wavefunctions
`.HSX`: Hamiltonian

Siesta ToolBox (STB)

STB-SUITE: A Companion to SIESTA

STB-SUITE is a Python toolkit that supports the SIESTA workflow.

It helps prepare calculation inputs, organize and validate run folders, and analyze results—while SIESTA performs the DFT calculations.

Install and Verify STB-SUITE



Install the engines and extras you need

SIESTA is installed separately. MACE tools need `pip install ".[ml]"`; optional dependencies report how to install them when missing.

A Workflow Is a Relay—Not a Black Box

One relaxed structure. Many calculations. One checked property.

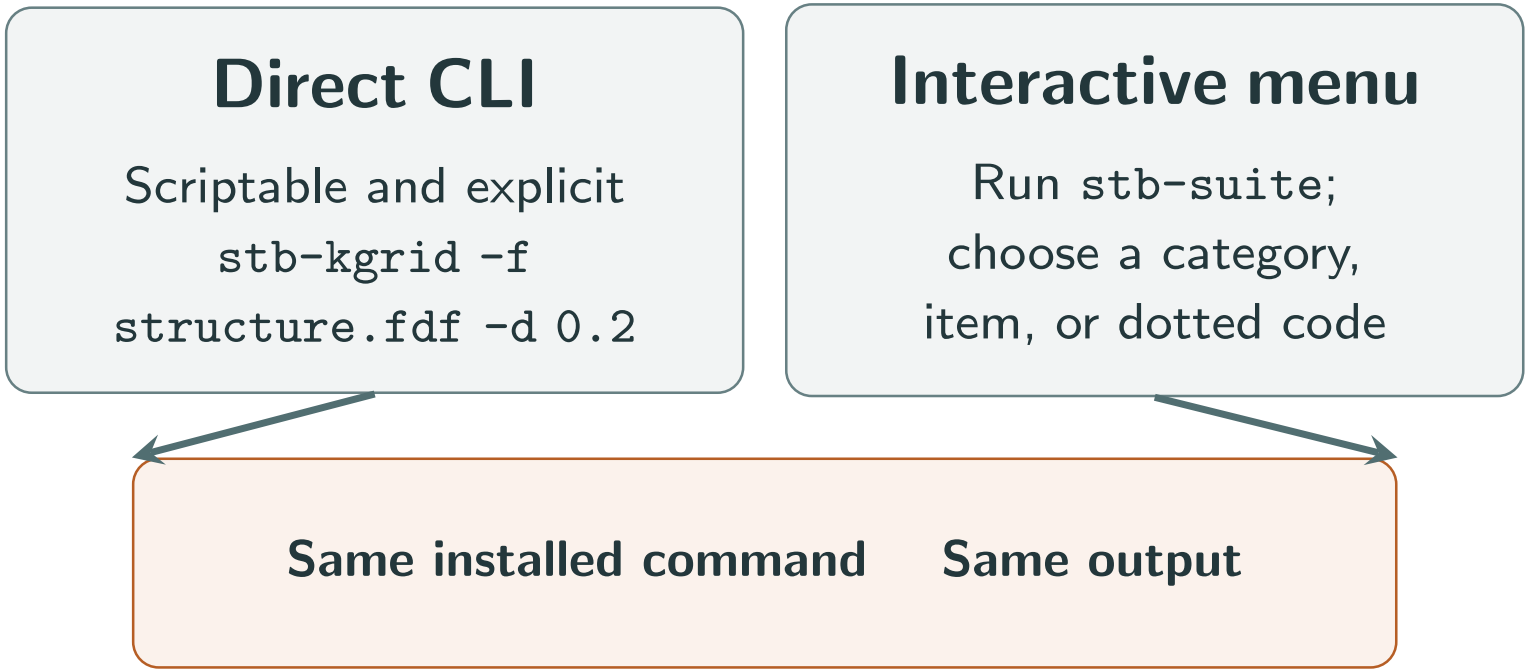


The handoff is the point

Analysis is a quality gate: unfinished or inconsistent runs are reported. Some methods loop back to **PREPARE** after an intermediate SIESTA run.

STB Manual, Chapter 5

Two Ways to Run a Tool



You control the calculation
STB does not choose how SIESTA is scheduled or launched; the menu and CLI are two interfaces to the same tools.

Reports, Diagnostics, and Citations

`[OK]`
Check passed

`[NOTE]`
Context or assumption

`[WARNING]`
Result needs attention

`[ERROR]`
Input refused

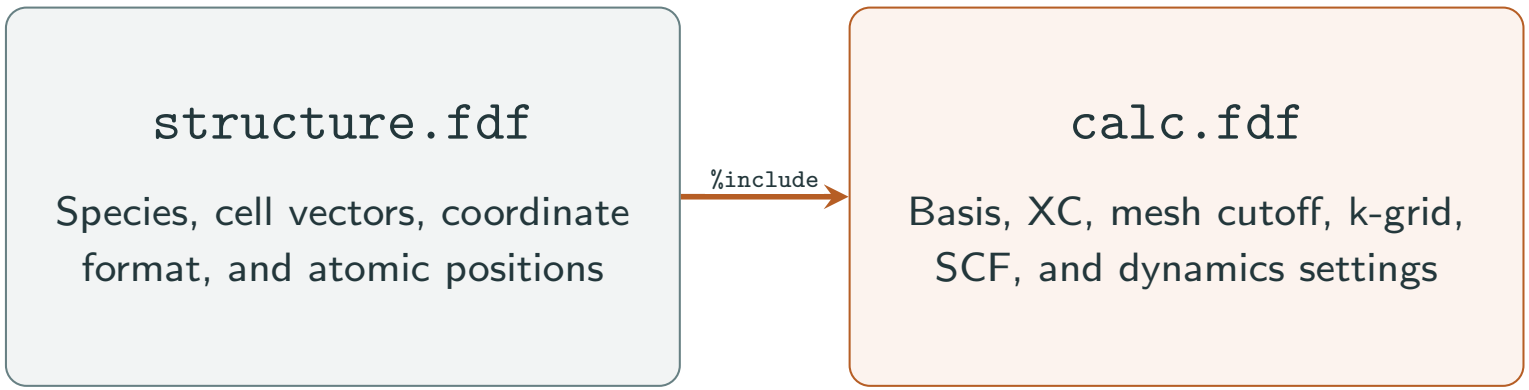
Read the full report

Numbered sections cover metadata, checks, results, and warnings. `--save-report` writes a copy; other markers include `[SKIPPED]`, `[UNVERIFIED]`, and `[EXTRAPOLATED]`.

Keep data and provenance

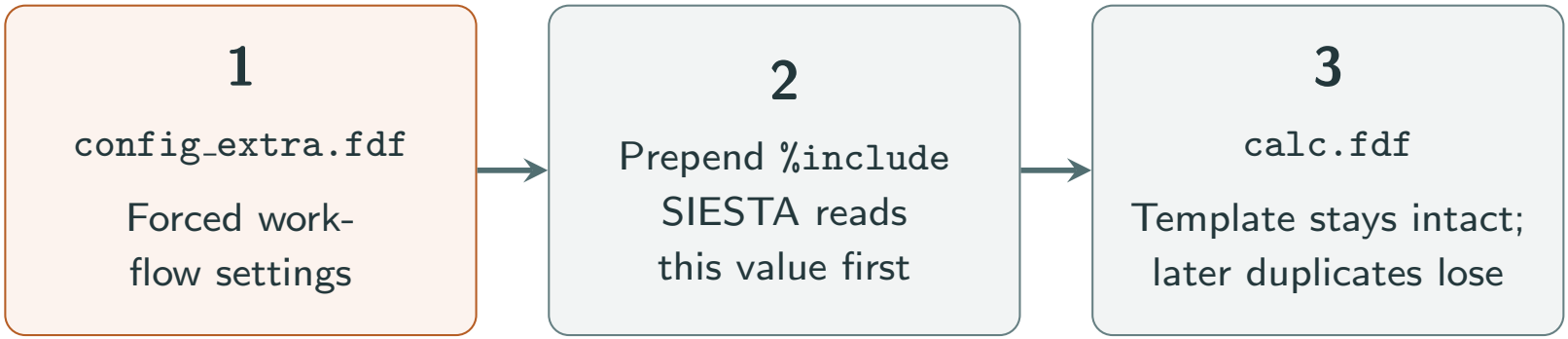
Tools merge citable sources into `references.bib`. Plots may save `.dat+.gplot`; trajectories support XSF, PDB, or extended XYZ.

Separate Geometry from Calculation Settings



Why keep them separate?
STB can prepare many geometries while copying your calculation template unchanged, so the intended level of theory stays explicit. A self-contained FDF is also supported by several tools.

Override Settings Without Editing Templates



Why order matters

FDF labels ignore case and punctuation; the first occurrence wins. An include appended later cannot override an existing template value.

Practical rule

Prepend the sidecar so forced settings take precedence without editing the user's calculation template.

Pseudopotentials: Sources and Validation

DOJO

PSML
72 elements
PseudoDojo
v0.5, PBE

VIRTUAL VAULT

PSF
69 elements
Troullier–Martins

YOUR BANK

Use a local folder
`stb-pseudo` resolves
and copies files

Portable calculation folders
STB copies pseudopotentials instead of symlinking them, so folders remain usable when moved or archived.

A database is not a quality guarantee
Validate a new pseudopotential against a known property for your system before production calculations.

Vacuum, Dimensionality, and Boundaries

0D	1D	2D	3D
Molecule	Wire	Slab	Bulk

How the dimensionality is found

A wrap-aware largest-gap test identifies vacuum axes (`--vacuum-gap`, default 10 Å); this affects k-grids and which properties or strains are meaningful.

Respect periodic boundaries

Coordinates are wrapped and distances use the minimum-image convention. Cell strains require fractional coordinates; Cartesian inputs are refused.

SIESTA Files Read by STB

RUN RECORD

`.out`: energy, Fermi level, forces, convergence
`.XV`: final geometry
`.FA`: forces

SPECTRA

`.bands`: bands
`.PDOS`: projected DOS
`.EIG`: eigenvalues

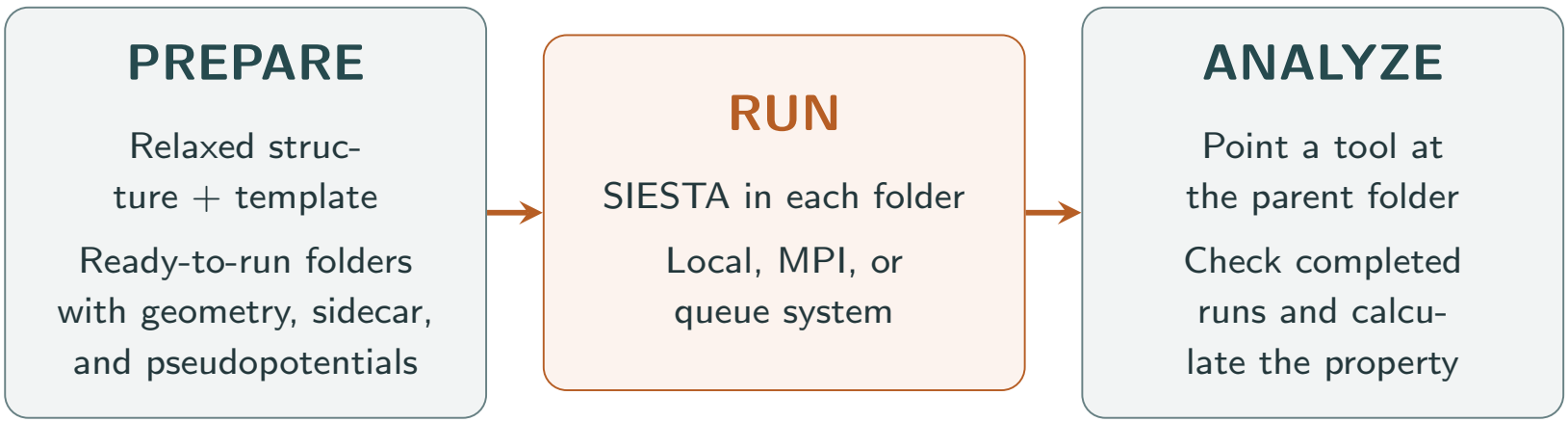
REAL SPACE

`.RHO`, `.VT`/`.VH`
`.LDOS`, wavefunctions, Hamiltonian

Match the file to the analysis

Some data require explicit SIESTA flags (for example `SaveRho`). STB can locate `calc.out` or accept an explicit path; filenames need not match `SystemLabel`.

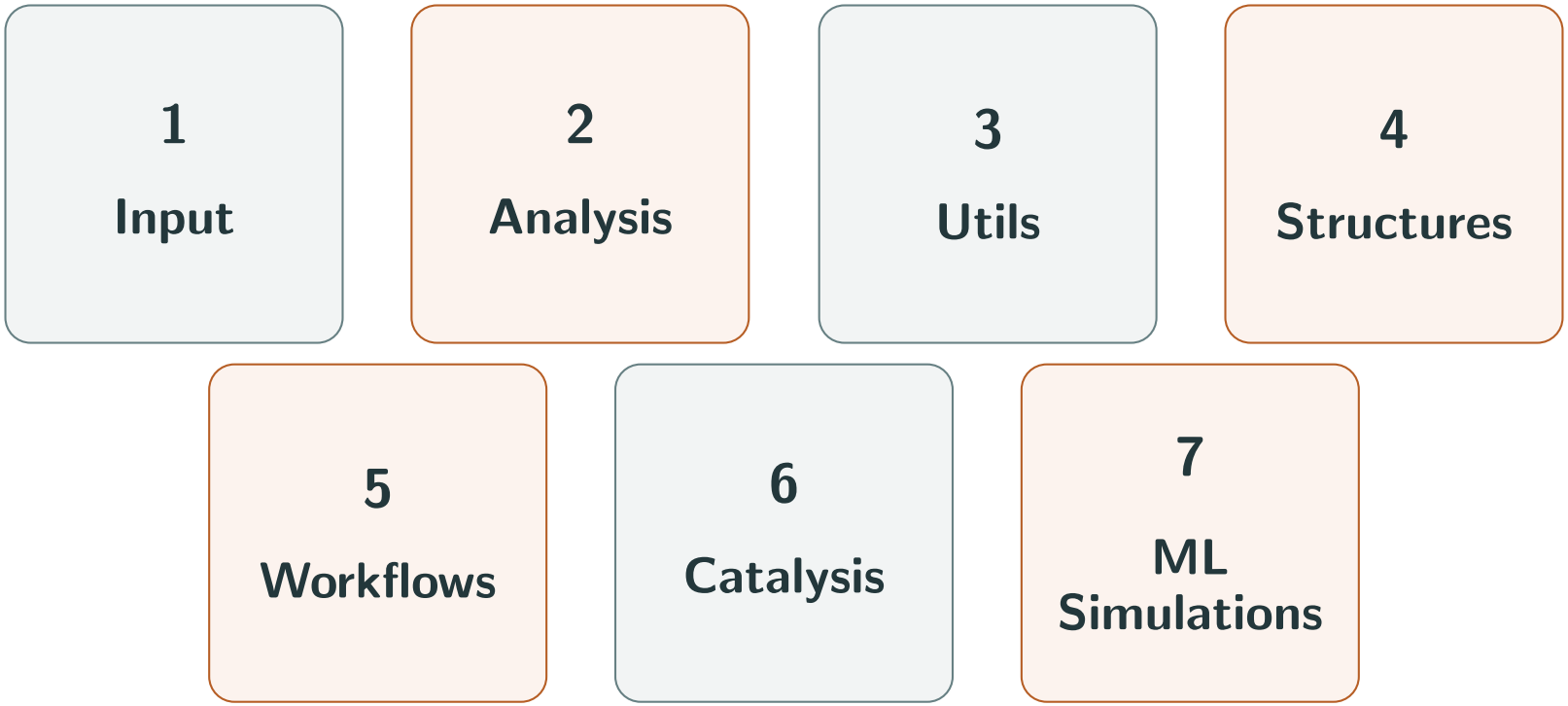
The Prepare–Run–Analyze Cycle



Some workflows need intermediate stages

Additional preparation may depend on results from an earlier SIESTA run; the same staged pattern still applies.

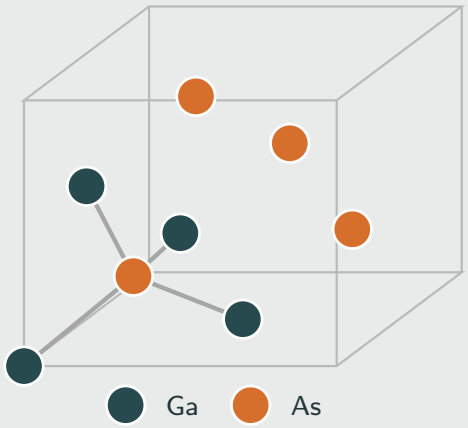
STB-SUITE Sections



Practical Part

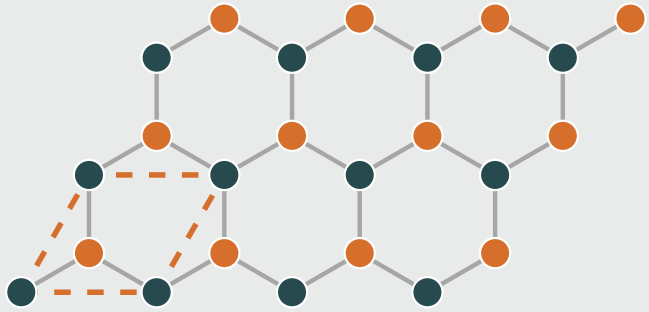
Practical Structures: GaAs and hBN

Zincblende GaAs



Cubic 3D lattice: interpenetrating fcc sublattices; each atom has four nearest neighbors of the other species.

Hexagonal Boron Nitride (hBN)



2D honeycomb lattice with a two-atom primitive cell; each B is bonded to three nearest-neighbor N atoms. Teal: B; orange: N.

GaAs primitive fcc vectors (cubic parameter a)

$$\mathbf{a}_1 = \tfrac{a}{2}(0,1,1)$$
$$\mathbf{a}_2 = \tfrac{a}{2}(1,0,1)$$
$$\mathbf{a}_3 = \tfrac{a}{2}(1,1,0)$$

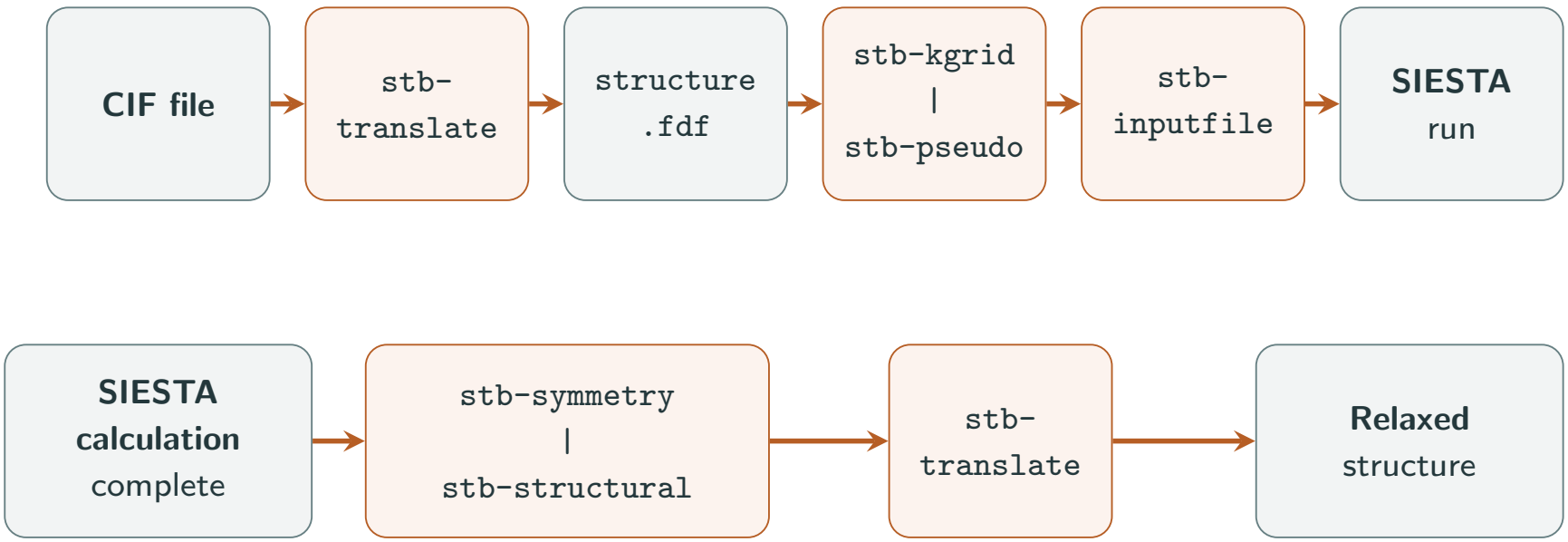
hBN primitive vectors (in-plane a)

$$\mathbf{a}_1 = a(1,0,0)$$
$$\mathbf{a}_2 = a\left(\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}, 0\right)$$

Crystallographic Information File (CIF)

A standard text format for crystal structures, storing unit-cell, symmetry, and atomic-position data.

Relaxation Workflow



Convergence Tests

Three Numerical Dials, Three Directions

Sweep the approximations before trusting the physics.

01 GRID

`Mesh.CutOff`

Real-space integration grid

$\longrightarrow$

Increase cutoff
finer grid

02 BASIS

`PAO.EnergyShift`

Orbital extent

$\longleftarrow$

Decrease shift
longer orbitals

03 K-GRID

Density-driven sampling

k-point sampling

$\longleftarrow$

Decrease density
denser grid

The sweeps are independent
Converge k-grid first for metals or small gaps; then recheck mesh and basis using that denser sampling.

STB Manual, Section 11.5

Stage 1: Sweep, Relax, Run

`stb-convergence` **prepares the test; you run SIESTA.**

INPUT

`structure.fdf` + production `calc.fdf`

Test the settings you plan to use

PREPARE

`stb-convergence`

One folder per swept value

Forced value in `config_extra.fdf`

RELAX

`value_1/` `value_2/` ...

Full CG relaxation Positions + variable cell

RUN

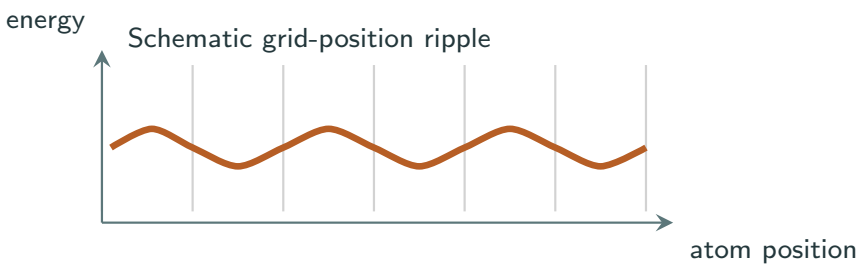
SIESTA in each folder

Local, MPI, or job array

You control the launch

Why relax every point?

The finite real-space grid creates an "eggbox" effect. At fixed geometry, energy may look settled while the relaxed cell is still drifting.

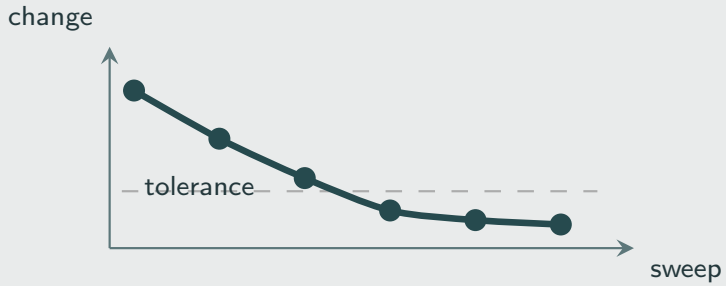


STB Manual, Section 11.5

Stage 2: Two Plateaus, One Recommendation

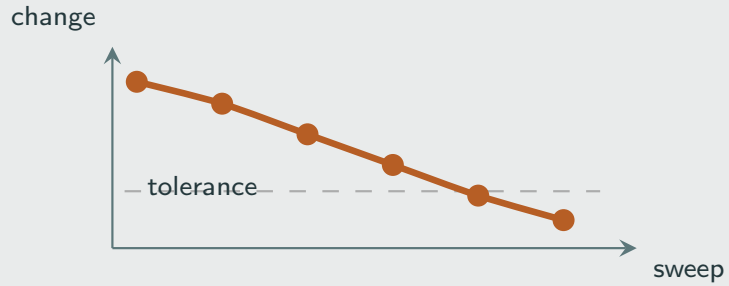
Energy and relaxed geometry must both converge.

Energy change



Default tolerance: 0.01 eV/atom

Relaxed-cell change



Default tolerance: 0.5% in volume

Schematic trends; each plot has its own scale.

Do not stop at the first flat-looking energy

Choose the more accurate value where both tests pass; the full remaining tail must stay within tolerance and SCF must be confirmed.

Quality flags are advisory, not plateau criteria.

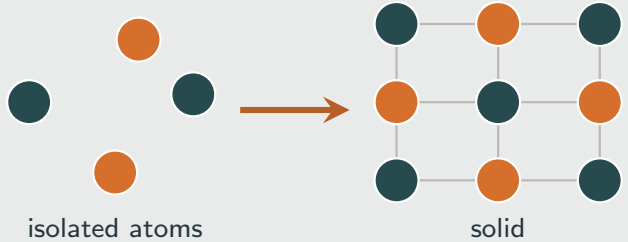
Optional `--apply` writes the recommendation to `calc.fdf`.

STB Manual, Section 11.5

Cohesive Energy

Cohesive Energy: Binding in a Solid

From free atoms to a crystal



Per-atom definition

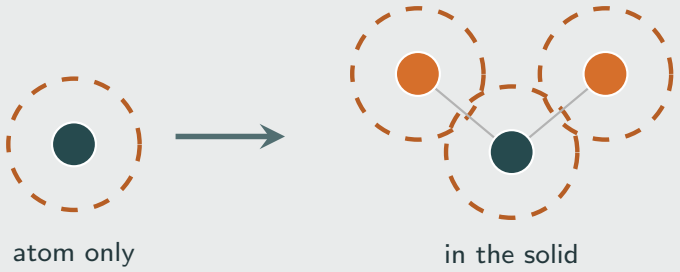
$$E_{\mathrm{coh}} = \frac{E_{\mathrm{bulk}} - \sum_{i=1}^{N} E_{\mathrm{atom}}(s_i)}{N}$$

- Crystal energy vs. the same atoms isolated.
- $E_{\mathrm{coh}} < 0$: bound; more negative means stronger binding.
- $-E_{\mathrm{coh}}$: energy per atom to separate the solid.
- Same basis and settings for bulk and atomic references.

STB Manual, Section 11.3

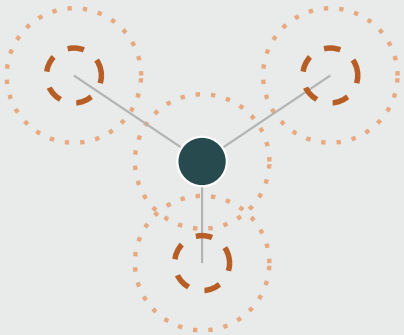
BSSE and Counterpoise: The Ghost-Atom Fix

Why localized bases overbind



- Localized LCAO/PAO basis functions.
- Neighbor orbitals lower the energy in the solid.
- Atomic reference too high $\Rightarrow$ E_{coh} too negative.

Counterpoise: ghost neighbors



- Ghosts add basis only: no nucleus or electrons.
- Recompute the atom with neighbors at crystal positions.
- `stb-cohesive` builds clusters; correction on by default.

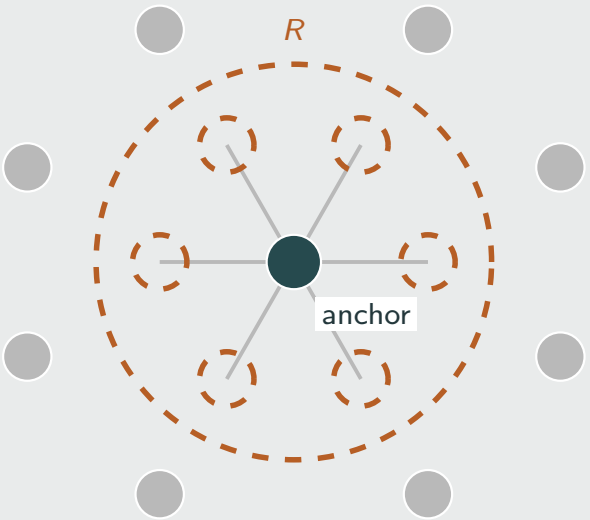
Counterpoise-corrected cohesive energy

$$E_{\mathrm{coh}}^{\mathrm{CP}} = \frac{E_{\mathrm{bulk}} - \sum_{i=1}^{N} E_{\mathrm{atom},i}^{\mathrm{ghost}}}{N}$$

STB Manual, Section 11.3

How STB Chooses Ghost Neighbors

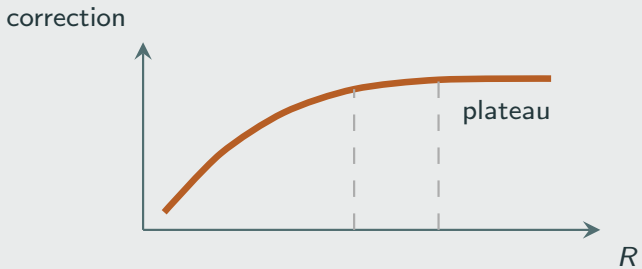
Neighbors within R become ghosts



Dashed: ghosts within R

Gray: outside R

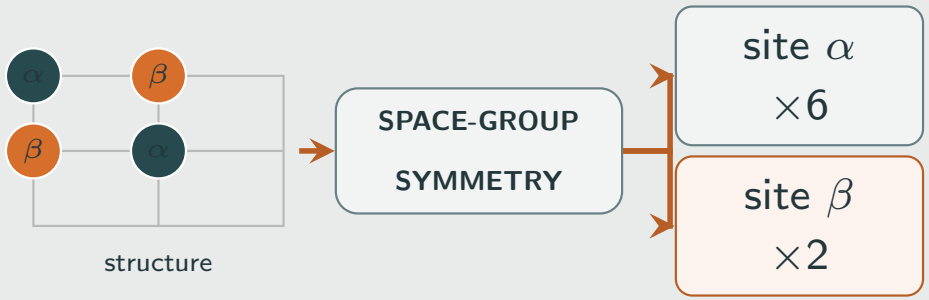
Tune and verify



- `--bsse-cutoff` R: default 4.0 Å.
- Shell count depends on bond length.
- Check `Ghost nb.`; zero means none included.
- `--bsse-convergence-check`: increase R; seek a plateau.

STB Manual, Section 11.3

Symmetry separates local environments

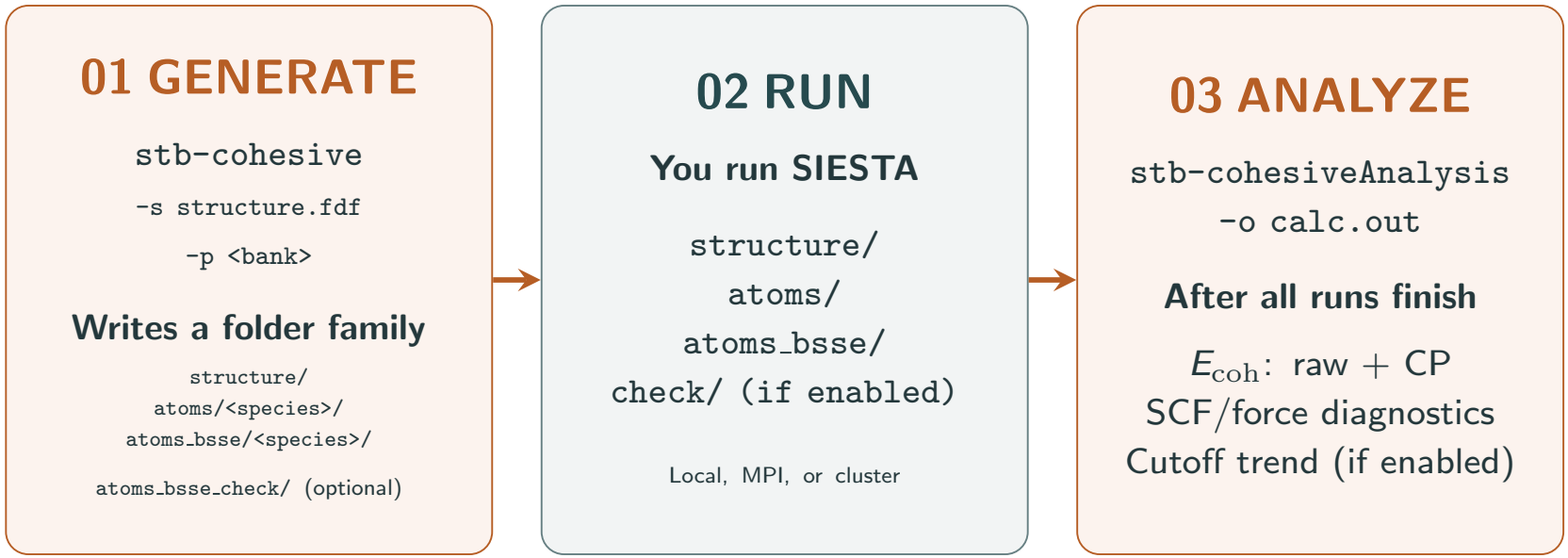


$$E_{\mathrm{ref}} = \frac{6E_\alpha + 2E_\beta}{8}$$

- One ghost cluster per inequivalent site.
- Weight references by site multiplicity, not a flat average.

STB Manual, Section 11.3

stb-cohesive: End-to-End Workflow



Key handoff

Stage 1 prepares folders; it does not launch SIESTA. Run every generated folder before analysis.

STB Manual, Section 11.3

Next ...

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Thank You & Let's Connect!

We're always open to new collaborations...

...especially if they involve discussing physics over a good cup of coffee!



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Thank you!